

Zeeshan Manzoor Bhat

CIRTA Postdoctoral Fellow | Structural & Earthquake Engineering

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RESEARCH INTERESTS

Large-scale experimental testing; Nonlinear modelling of structures; Seismic hazard, risk and vulnerability assessment; Performance-based earthquake engineering; Resilience-based seismic design.

EDUCATION

Ph.D. in Structural and Earthquake Engineering

July 2018 – March 2025

Indian Institute of Technology Roorkee, India

Thesis: *Seismic Performance of Masonry Infilled RC Frame Buildings Designed for Modern Codes*

Advisors: [Prof. Yogendra Singh](#) & [Prof. Pankaj Agarwal](#)

M.Tech. in Structural Engineering

July 2016 – July 2018

National Institute of Technology Srinagar, India

Thesis: *An Experimental Investigation into the Behavior of Steel–Timber Composite Beams*

Advisor: [Prof. Javed Ahmad Bhat](#)

B.Tech. in Civil Engineering

July 2012 – June 2016

National Institute of Technology Srinagar, India

RESEARCH EXPERIENCE

CITRA Postdoctoral Fellow

August 2026 – Ongoing

Queen's University, Kingston, Canada

Project: *Next-Generation Climate-Resilient and Energy-Efficient Housing*

Research Associate

June 2025 – June 2026

Indian Institute of Technology Roorkee, India

Project: *Enhancing Sustainability of RC Frame Buildings through Seismic Safety and Design Life Elongation Using GFRP Reinforcement*

FUNDED PROJECTS

Next-Generation Climate-Resilient and Energy-Efficient Housing

Queen's University, Kingston, Canada

Funding: \$140,000 through [Canada Impact + Research Training Awards](#)

Supervisor: [Prof. Amir Fam](#)

PUBLICATIONS

Peer-Reviewed Journal Articles

- [1] Bhat, Z. M., & Singh, Y. (2026). Out-of-plane seismic fragility analysis of masonry infills in low- and mid-rise RC frame archetype buildings accounting for in-plane damage. *Journal of Building Engineering*. [Link](#)
- [2] Bhat, Z. M., & Singh, Y. (2025). Seismic risk assessment of fly ash brick and AAC block masonry infilled RC frame buildings designed for modern codes. *Journal of Earthquake Engineering*. [Link](#)
- [3] Bhat, Z. M., & Singh, Y. (2025). Out-of-plane behavior of masonry infills of different types and slenderness ratios under reversed cyclic loading. *Journal of Structural Engineering*. [Link](#)
- [4] Bhat, Z. M., Singh, Y., & Agarwal, P. (2024). Cyclic testing and diagonal strut modelling of different types of masonry infills in reinforced concrete frames designed for modern codes. *Engineering Structures*. [Link](#)

- [5] Bhat, Z. M., Singh, Y., & Agarwal, P. (2023). Characterization of mechanical behavior of different types of masonry with full-field strain using digital image correlation. *Construction and Building Materials*. [Link](#)
- [6] Waseem, S. A., Bhat, Z. M., & Bhat, J. A. (2022). Experimental investigation into the behavior of steel–timber composite beams. *Practice Periodical on Structural Design and Construction*. [Link](#)

Under Review

- [7] Bhat, Z. M., & Singh, Y. Full-scale cyclic testing and DIC-based damage assessment of masonry infilled RC frames with openings. *ACI Structural Journal*.

INTERNATIONAL CONFERENCES

- [1] Bhat, Z. M., & Singh, Y. (2025). Reversed out-of-plane cyclic testing of masonry infills using double-leaf airbags. *COMPDYN 2025: 10th International Conference on Computational Methods in Structural Dynamics and Earthquake Engineering*, Rhodes Island, Greece, 15–18 June 2025. [Link](#)
- [2] Bhat, Z. M., & Singh, Y. (2025). Cyclic in-plane behavior of masonry infilled RC frames with detailed interpretation of damage using digital image correlation. *15th Canadian Masonry Symposium*, Ottawa, Canada, 2–5 June 2025. [Link](#)

TECHNICAL REPORTS

- [1] Bhat, Z. M., Singh, Y., & Agarwal, P. (2024). *Seismic Safety Measures for RC Frame Buildings with Different Types of Infill Panels*. NBCC (India) Limited, India.
- [2] Team Member. (2022). *Compendium of Traditional Earthquake-Resilient Construction Practices*. NDMA, Government of India. [Link](#)

OPEN DATASETS

- [1] Bhat, Z. M., & Singh, Y. (2024). Fly ash brick and AAC block masonry infilled RC frame buildings: OpenSees models and responses. *DesignSafe-CI*. [Link](#)

AWARDS AND ACHIEVEMENTS

- Canada Impact + Research Training Award – Postdoctoral, **Natural Sciences and Engineering Research Council of Canada (NSERC)**, 2026
- Research Fellowship, **Sapienza University of Rome, Italy**, 2025
- MHRD Fellowship for Doctoral Studies, **Ministry of Human Resource Development, Government of India**
- MHRD Fellowship for M.Tech Studies, **Ministry of Human Resource Development, Government of India**
- Ranked among the top 2% in the **Graduate Aptitude Test in Engineering (GATE)** 2016, among 119,873 candidates

TECHNICAL SKILLS

- **Programming & Tools:** Python, MATLAB, TCL, LaTeX
- **Software:** OpenSees, STKO, ETABS, SAP2000, OpenQuake
- **Experimental Techniques:** Digital Image Correlation (DIC), Laboratory Instrumentation and Data Acquisition, Servo-hydraulic Actuator Control, Load Cell Calibration

INTERNATIONAL WORKSHOPS

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|---|-----------------|
| 6th International Course on Seismic Analysis of Structures | 19–22 July 2021 |
| <i>University of Palermo, Italy</i> | |
| Nonlinear Modelling of RC Structures Using OpenSees | 22–26 June 2020 |
| <i>Università degli Studi di Napoli Federico II, Italy</i> | |

TEACHING ASSISTANTSHIPS

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|---|-------------|
| Advanced Earthquake Resistant Design of Structures (EQN-512) | IIT Roorkee |
| Instructor: Prof. Yogendra Singh | |
| Earthquake Resistant Design of Structures (EQN-563) | IIT Roorkee |
| Instructor: Prof. Yogendra Singh | |

REFERENCES

Prof. Yogendra Singh

Professor, Department of Earthquake Engineering, Indian Institute of Technology Roorkee, India

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Prof. Amir Fam

Professor, Department of Civil Engineering, Queen's University Kingston, Canada

Email: amir.fam@queensu.ca | Phone: +1-613 985 3694 | [Website](#) | [Google Scholar](#)

Prof. Pankaj Agarwal

Professor, Department of Earthquake Engineering, Indian Institute of Technology Roorkee, India

Email: pankaj.agrawal@eq.iitr.ac.in | Phone: +91-9837004618 | [Website](#) | [Google Scholar](#)